Exceptional Cases and Statistical Structure of Sequence A247975: a Computational Study of Sun's Conjecture 4.1(i)

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Abstract

We present a computational investigation of sequence A247975 in the OEIS, which encodes Sun's Conjecture 4.1(i): for every positive integer m, there exists a positive integer n such that $(m+n)$ divides $p_m^2+p_n^2$, where p_k denotes the k-th prime. We determine $a(m)$ for $119\,995$ of the values $m = 1, 2, \ldots, 120\,000$, with final search bound $n \leq 2 \times 10^{11}$; the remaining 5 values are unresolved within this bound. We prove that the modular filter used in the search eliminates an asymptotically total proportion of candidates, and establish computational lower bounds $a(m) > 2\times 10^{11}$ for the five unresolved cases. Among the resolved values we identify 606 exceptional cases (where $a(m) > 10^8$), including the outstanding Cramér-type ratio $a(m)/(m \log m) \approx 196\,577$ at $m = 23\,995$ and the largest absolute value $a(82\,300) = 81\,961\,045\,941$. Statistical analysis yields a Hill statistic $\hat{\alpha}_{\text{Hill}} \approx 0.713$, consistent with heavy-tailed behaviour; a density-based heuristic model fails quantitatively, and we identify three structural sources of discrepancy. The five unresolved values are proposed as computational challenge instances. Divisibility was independently verified for all 606 exceptional cases via `SymPy`; minimality was independently verified for 319 cases via an independent C sieve.

Keywords: Prime numbers, Integer sequences, OEIS, Computational number theory, Sun's conjecture, Cramér ratio

MSC Classification: 11A41 , 11Y55 , 11N05

1 Introduction

In a paper submitted to arXiv in 2013 and published in 2019, Zhi-Wei Sun [1] proposed a collection of conjectures on addition and divisibility involving primes and other arithmetic functions. Among these, Conjecture 4.1(i) (p. 15) states the following.

Conjecture 1.1 (Sun, 2013/2019) *For every positive integer m, there exists a positive integer n such that*

$$(m+n) \mid p_m^2 + p_n^2,$$

where p_k denotes the k-th prime.

The conjecture was submitted to the OEIS in September 2014 and appears as sequence A247975 [2]. A closely related sequence, A248354 [3], encodes the analogous conjecture for $p_{m^2}^2 + p_{n^2}^2$ (Conjecture 4.1(ii) in [1]). The sequence $a(m)$, defined as the least positive integer n satisfying the divisibility condition, was computed for $m \leq 10\,000$ by Chai Wah Wu [12]. To the best of our knowledge, no other published computational extension of sequence A247975 beyond Wu's $m \leq 10\,000$ exists in the literature.

The present work has three main contributions. First, we extend the computation to $m = 1, \ldots, 120\,000$, providing the largest verified dataset for this conjecture to date. Second, we carry out a systematic statistical analysis of the distribution of $a(m)$. Third, we identify and characterize 5 values of m for which the conjecture resists verification within our final search bound $n \leq 2 \times 10^{11}$, all exhibiting elevated modular obstruction rates and extreme Cramér-ratio lower bounds; these are proposed as explicit open problems.

2 Setup and Notation

We reformulate the divisibility condition in modular arithmetic and establish the number-theoretic constraints underlying the computational search.

For positive integers m and n, set $s = m + n$. The divisibility condition $(m+n) \mid p_m^2 + p_n^2$ is equivalent to

$$p_n^2 \equiv -p_m^2 \pmod{s}.$$

When $\gcd(p_m, s) = 1$, this simplifies to

$$\left(p_n p_m^{-1}\right)^2 \equiv -1 \pmod{s},$$

so (under the assumption $\gcd(p_m, s) = 1$) a solution exists at index n if and only if p_n falls in one of the admissible residue classes of the form $r \cdot p_m \pmod{s}$, where r ranges over the square roots of -1 modulo s and $\omega(s)$ denotes the number of distinct prime factors of s. When such square roots exist, their number is $2^{\omega(s)}$ for odd s and $2^{\omega(s)-1}$ for $s \equiv 2 \pmod{4}$ (see below).

When $\gcd(p_m, s) > 1$, i.e., $p_m \mid s = m + n$, the divisibility chain $p_m \mid s$, $p_m \mid p_m^2$, and $(m+n) \mid (p_m^2 + p_n^2)$ together force $p_m \mid p_n^2$, hence $p_m \mid p_n$ (since p_m is prime), hence

$p_m = p_n$ (since both are prime), so $n = m$. In this branch the *only* possible solution at modulus s is $n = m$, and this case is excluded for $m > 1$ by the following argument.

If $n = m$ were a solution it would require $2m \mid 2p_m^2$, i.e., $m \mid p_m^2$. Since p_m is prime, $m \mid p_m^2$ implies $m \in \{1, p_m, p_m^2\}$. The case $m = p_m^2$ is impossible for $m \geq 2$: since $p_m \geq 2$ we have $p_m^2 \geq 2p_m > 2m > m$, a contradiction. The case $m = p_m$ is ruled out by an elementary observation: since $p_1 = 2, p_2 = 3, \ldots$ is a strictly increasing sequence of integers starting at 2, we have $p_m \geq m + 1 > m$ for all $m \geq 1$, so $p_m \neq m$. Hence $n = m$ is not a valid solution for any $m \geq 2$. For $m = 1$, by contrast, $n = m = 1$ is a valid solution: $m + n = 2$ divides $p_1^2 + p_1^2 = 8$, so $a(1) = 1$. The algorithm handles this edge case before the main search. For $m \geq 2$, the $n = m$ branch is excluded and the algorithm proceeds to the main search loop.

A classical result (see [5], Ch. 5) gives:

Proposition 2.1 *-1 is a quadratic residue modulo s if and only if both of the following hold:*

(a) $4 \nmid s$ (equivalently, $v_2(s) \leq 1$, where $v_2(s)$ denotes the 2-adic valuation of s), and
(b) no prime $q \equiv 3 \pmod 4$ divides s.

Remark 2.2 Condition (b) is strictly stronger than requiring the Jacobi symbol $\left(\frac{-1}{s}\right) = 1$. For odd s, the Jacobi symbol $\left(\frac{-1}{s}\right) = 1$ if and only if the *sum* $E = \sum_{q \equiv 3 \pmod 4} v_q(s)$ of the exponents of all primes $q \equiv 3 \pmod 4$ dividing s is even (including $E = 0$). By contrast, actual solvability of $x^2 \equiv -1 \pmod s$ requires that no such prime divides s at all, i.e., $E = 0$. Thus the Jacobi symbol can equal 1 even when -1 is *not* a quadratic residue modulo s; a simple example is $s = 21 = 3 \times 7$, for which $E = 1 + 1 = 2$ is even so $\left(\frac{-1}{21}\right) = 1$, yet -1 is not a quadratic residue modulo 21 because condition (b) fails. The Jacobi symbol is therefore only a necessary condition for solvability, not a sufficient one. For $s \equiv 2 \pmod 4$, the Jacobi symbol is undefined (it is only defined for odd positive denominators); solvability of $x^2 \equiv -1 \pmod s$ in this case requires that no odd prime $q \equiv 3 \pmod 4$ divides s, which is exactly condition (b) applied to the odd part of s.

When -1 is a QR mod s and s is odd (so every prime divisor of s is $\equiv 1 \pmod 4$), there are exactly $2^{\omega(s)}$ square roots of -1 modulo s, where $\omega(s)$ denotes the number of distinct prime factors of s (see [5], Ch. 5, Theorem 9). When $s \equiv 2 \pmod 4$ (i.e., $v_2(s) = 1$), condition (a) is satisfied and the factor of 2 contributes exactly one solution to $x^2 \equiv -1 \pmod 2$, so the total count is $2^{\omega(s)-1}$; this case is handled correctly by the filter (see Section 3.1).

The modular filter in our implementation checks condition (a) exactly and condition (b) partially (up to a trial-division bound); see Section 3.1 for details. Candidates not eliminated by the filter are tested by direct arithmetic.

3 Computational Methods

3.1 Algorithm and modular filter

Our implementation (written in C) uses a segmented sieve of Eratosthenes [14] to build a table of all primes up to a specified limit, together with a modular filter to

skip values of n for which $s = m + n$ admits no square root of -1, before testing the divisibility condition directly.

The filter checks, for each candidate n, two conditions:

(a) $4 \nmid s$, i.e., $s \not\equiv 0 \pmod 4$ (condition (a) of Proposition 2.1, checked exactly); and
(b′) no prime $q \equiv 3 \pmod 4$ up to a trial-division bound $B = 10^4$ divides s (a partial check of condition (b) of Proposition 2.1).

This is a safe filter: it eliminates only values of s for which $x^2 \equiv -1 \pmod s$ provably has no solution, so no valid candidate is ever discarded. More precisely, if either condition (a) or (b) of Proposition 2.1 is violated for s, then no $x \in \mathbb{Z}$ satisfies $x^2 \equiv -1 \pmod s$, and consequently the divisibility $(m + n) \mid p_m^2 + p_n^2$ cannot hold when $\gcd(p_m, s) = 1$ (since it would require $(p_n p_m^{-1})^2 \equiv -1 \pmod s$); the filter is therefore *sound* in the sense that it discards only provably impossible candidates. Note that for $m \geq 2$ the case $\gcd(p_m, s) > 1$ would force $n = m$, which is not a valid solution (as shown in Section 2); the filter may therefore assume $\gcd(p_m, s) = 1$ for all remaining candidates, and its conditions correctly characterise impossibility. Values of s that have a prime factor $q \equiv 3 \pmod 4$ larger than B are not eliminated by the filter; they pass through to the direct divisibility test, which may still fail, resulting in a small amount of redundant arithmetic. Across the range $m = 1, \ldots, 120\,000$, the filter eliminates empirically 73–77% of candidate values of n before any arithmetic is performed; we call this per-m elimination percentage the *obstruction rate*. The following proposition makes the asymptotic obstruction rate precise.

Proposition 3.1 (Asymptotic obstruction rate) *Let $\mathcal{A}(x)$ denote the set of integers $s \leq x$ satisfying both conditions of Proposition 2.1, i.e., $v_2(s) \leq 1$ and s has no prime factor $q \equiv 3 \pmod 4$. Then*

$$\frac{|\mathcal{A}(x)|}{x} \sim \frac{C_0}{\sqrt{\ln x}} \qquad (x \to \infty),$$

where $C_0 > 0$ is an explicit constant. In particular, the proportion of candidates eliminated by the exact filter (conditions (a) and (b) of Proposition 2.1) tends to 1 as the search bound grows.

Proof An integer s satisfies condition (b) if and only if every prime factor of s lies in the set $\mathcal{P} = \{2\} \cup \{p \text{ prime} : p \equiv 1 \pmod 4\}$. The classical B_1 numbers (integers representable as a sum of two squares) are characterised by the condition that every prime $q \equiv 3 \pmod 4$ appears to an *even* exponent in the factorisation; our set $\mathcal{A}(x)$ imposes the strictly stronger requirement that no such prime divides s at all, so $\mathcal{A}(x)$ is a proper subset of the B_1 set. By the prime number theorem for arithmetic progressions, the primes in $\mathcal{P}$ have natural density $\frac{1}{2}$ among all primes. A theorem of Wirsing [17] on sums of non-negative multiplicative functions gives the count of integers up to x whose prime factors all belong to a set of primes with density δ as $\sim C \cdot x/(\ln x)^{1-\delta}$ for an explicit constant $C > 0$ depending on the set. Applying this with $\delta = \frac{1}{2}$ yields $|\{s \leq x : \text{condition (b) holds}\}| \sim C_1 \cdot x/\sqrt{\ln x}$; the asymptotic order matches the classical Landau–Ramanujan result [15] for the larger B_1 set, but the constant C_1 is smaller since fewer integers are counted. Imposing condition (a) additionally ($v_2(s) \leq 1$) excludes those s divisible by 4. Among integers whose prime factors lie in $\mathcal{P}$, the proportion with $v_2(s) \geq 2$ is asymptotically constant, so the combined count retains the same order

$\sim C_0 \cdot x/\sqrt{\ln x}$ with a reduced constant $C_0 \leq C_1$. The claimed obstruction rate follows: $1 - |\mathcal{A}(x)|/x \to 1$ as $x \to \infty$. □

Remark 3.2 The constant C_0 in Proposition 3.1 can be estimated numerically via a Mertens-type product. Define

$$C_3 = \lim_{y\to\infty} \sqrt{\ln y} \prod_{\substack{q\leq y \\ q\equiv 3 \pmod 4}} \left(1 - \frac{1}{q}\right),$$

which exists by the analogue of Mertens' third theorem for primes in arithmetic progressions (see [10], Ch. I.3). Computing the partial products up to $y = 10^6$ gives $C_3 \approx 0.8689$, stable to six decimal places. Combining with the factor $\frac{3}{4}$ from condition (a) yields $C_0 \approx \frac{3}{4} \times 0.8689 \approx 0.652$.

For the *truncated* filter used in our implementation, which checks condition (b) only up to a trial-division bound $B = 10^4$, the survival proportion is

$$\frac{3}{4} \prod_{\substack{q\leq B \\ q\equiv 3 \pmod 4}} \left(1 - \frac{1}{q}\right) \approx 0.215,$$

yielding a predicted obstruction rate of $\approx 78.5\%$. This is consistent with, though slightly higher than, the observed range 73–77%; the residual gap reflects correlations among consecutive values $s = m + n$ arising from shared small prime factors and the arithmetic structure of the sequence $\{m + 1, m + 2, \ldots\}$. We refer to this per-m elimination percentage as the *obstruction rate*; see Section 7 and Table 5 for the obstruction rates of the unresolved cases.

All integer arithmetic uses 64-bit signed integer types (`int64_t`) throughout the main computation (Phase 6). The divisibility check $(p_m^2 + p_n^2) \bmod s = 0$ is implemented using explicit reduction before multiplication:

$$\big((p_m^2 \bmod s) + (p_n \bmod s)^2\big) \bmod s = 0.$$

The formula treats p_m and p_n asymmetrically for a concrete reason: for $m \leq 120\,000$ we have $p_m \leq 1\,583\,539$ (the exact value of $p_{120\,000}$), so $p_m^2 \leq 2.51 \times 10^{12}$ fits directly in `int64_t` and is reduced after squaring; by contrast, p_n can reach 4.71×10^{10} (at $n = 2 \times 10^9$), so $p_n^2 \approx 2.2 \times 10^{21}$ would overflow `int64_t` unless p_n is reduced *before* squaring. After reduction, $\texttt{pnr} = p_n \bmod s < s \leq m + n \leq 2.0001 \times 10^9$, and hence

$$\texttt{pnr}^2 < (2.0001 \times 10^9)^2 = 4.0004 \times 10^{18} < 2^{63} - 1 \approx 9.22 \times 10^{18}.$$

To close the overflow argument: the sum $(p_m^2 \bmod s) + \texttt{pnr}^2$ satisfies $(p_m^2 \bmod s) \leq s - 1 < 2.0001 \times 10^9$ and $\texttt{pnr}^2 \leq 4.0004 \times 10^{18}$, so the sum is at most $4.0004 \times 10^{18} + 2.0001 \times 10^9 < 2^{63} - 1$, and the final reduction is safe. No overflow occurs on `int64_t`; neither `__int128` nor arbitrary-precision arithmetic is required.

In the extended search (Section 3.3), the search bound reaches $n \leq 2 \times 10^{11}$, so $s = m+n$ can reach $\approx 2\times10^{11}$ and the product $p_n^2 \bmod s$ requires intermediate values that may exceed 2^{63}. Accordingly, the extended-search program uses `__uint128_t` for the modular squaring step $p_n^2 \bmod s$ throughout; peak RAM is kept below 100 MB by replacing the global prime table with a segmented sieve that computes primes on the fly; see Section 3.3 for details.

Prime generation uses a bitset segmented sieve of Eratosthenes storing one bit per odd number, which halves memory relative to a full-integer representation and is fully deterministic: every number up to the sieve limit is either confirmed prime or confirmed composite; no probabilistic primality test is employed.

3.2 Computation phases

The computation proceeded in phases:

1. *Phases 1–2* ($m \leq 10\,000$, bound $n \leq 10^8$): established baseline results, resolving 9998 of 10 000 cases.
2. *Phase 3* ($m \leq 10\,000$, bound $n \leq 7.6 \times 10^8$): resolved the remaining 2 cases, confirming $a(4703) = 760\,027\,770$ (previously noted by Sun [1]) and establishing $a(5263) = 420\,423\,462$.
3. *Phases 4–6* ($m \leq 120\,000$, bound $n \leq 2\times10^9$): extended the dataset to $m = 120\,000$, resolving 119 931 of 120 000 cases, with 69 values unresolved at the Phase 6 bound.

All computations were performed on a personal workstation equipped with an AMD Ryzen 9 7940HS processor and 64 GB of RAM. The final phase (Phase 6, $m \leq 120\,000$) required approximately 9 minutes of wall-clock time and approximately 20.8 GB of RAM: $\sim$ 3.2 GB for the packed-bit sieve up to 5.14×10^{10} (one bit per odd number) and $\sim$ 17.6 GB for the prime index table storing all 2 176 824 767 primes below 5.14×10^{10} as 8-byte integers. The search bound $n \leq 2 \times 10^9$ corresponds to $p_n \leq 47\,055\,833\,459 \approx 4.71\times10^{10}$. The sieve is built to 5.14×10^{10}, a margin of approximately 9% above $p_{2\times10^9}$, to provide a comfortable buffer that avoids edge effects and ensures all needed primes are available.

3.3 Extended search

After the initial computation (Phase 6) left 69 values unresolved at bound $n \leq 2\times10^9$, a targeted extended search progressively extended the bound to $n \leq 2\times10^{11}$, resolving 64 of the 69 cases and reducing the unresolved count to 5.

The extended search used a new program (`sun_ext_v11.c`) with a fundamentally different architecture from the Phase-6 program: instead of building a global prime table (which would require $\sim$ 20 GB of RAM at bound 10^{10}, and far more at 2×10^{11}), it uses a segmented sieve with no persistent prime table, computing primes on the fly within segments of 4×10^6 odd slots (0.5 MB per thread). The starting prime $p_{\text{START_N}}$ is computed once; all subsequent primes are enumerated within segments. Peak RAM stays below 100 MB regardless of the search bound. Parallelism is achieved via OpenMP on 16 threads.

The modular squaring step uses a 128-bit integer type (`__uint128_t`) for $p_n^2 \bmod s$ throughout the extended search (where p_n^2 may exceed 2^{63}); a single `DIVQ` inline-assembly instruction handles the final modular reduction efficiently.

Table 1 summarises the extended search. The search was halted after Session 4 ($n \leq 2 \times 10^{11}$), at which point two consecutive sessions (S3 and S4) had each resolved zero additional cases, strongly suggesting that the remaining 5 unresolved values likely require n well beyond 2×10^{11}.

Table 1 Extended search: progressive resolution of the 69 cases unresolved after the initial computation (Phase 6). "+Resolved" denotes newly resolved cases in each session.

Session	Bound n	+Resolved	Still open	Time (min)	Code
Phase 6 (initial)	$\leq 2 \times 10^9$	—	69	8.8	`sun_v6.c`
S1	$\leq 5 \times 10^{10}$	62	7	49.0	`sun_ext_v11.c`
S2	$\leq 10^{11}$	2	5	68.8	`sun_ext_v11.c`
S3	$\leq 1.5 \times 10^{11}$	0	5	102.8	`sun_ext_v11.c`
S4	$\leq 2 \times 10^{11}$	0	5	135.1	`sun_ext_v11.c`

The total wall-clock time for Sessions S1–S4 was approximately 356 minutes ($\approx$ 6 hours).

3.4 Use of AI tools

The computational implementation, including the C search program and verification sieve, was developed with extensive assistance from Anthropic's Claude language models (`claude-sonnet-4-6` and `claude-opus-4-6`). Claude also assisted substantially in the statistical analysis and in the preparation of this manuscript. All computations were executed and independently verified by the author, who bears full scientific responsibility for all results, including those portions where AI assistance was used.

3.5 Independent verification

Results were verified by three methods, as described below. Minimality of $a(m)$ is fully independently verified for 319 values; for all other values it depends on the correctness of the exhaustive C search.

1. *Cross-consistency*: all 10 000 values for $m \leq 10\,000$ agree exactly between Phase 3 and Phase 6 outputs, confirming implementation stability and determinism across independent runs. Note that this method confirms reproducibility of the algorithm, not its correctness; two runs of the same implementation will agree regardless of any systematic bug.
2. *Direct divisibility via* `SymPy`: for each of the 606 exceptional resolved cases ($a(m) > 10^8$), the values p_m and $p_{a(m)}$ were computed independently using `sympy.prime()` [7], which implements its own prime-generation algorithm entirely separate from the C sieve, and we verified $(p_m^2 + p_n^2) \bmod (m+n) = 0$ directly. All 606 verified without error: 542 cases from the initial computation and all 64 cases

from the extended search. For the 64 extended-search cases, $p_{a(m)}$ values range from $\approx 4.8 \times 10^{10}$ to $\approx 2.2 \times 10^{12}$; these were computed via `sympy.primepi()` combined with a walk using `sympy.nextprime()`/`prevprime()`, starting from a PNT estimate. The `SymPy` and Python scripts take as input only the values of m and the claimed $a(m)$; they do not use the prime table produced by the C program. This method confirms the divisibility condition for the claimed values but *not* minimality; for exceptional cases not covered by Method 3 below, minimality depends on the correctness of the C implementation.

3. *Independent sieve*: an independent C implementation (`verify_min_ext.c`) using a segmented sieve of Eratosthenes, with no code shared with the search programs, verified 319 cases by exhaustive search from $n = 1$, confirming both the divisibility condition *and* the minimality of $a(m)$. The 319 cases comprise six groups (318 cases total) plus one smoke-test reference case:

   - *Group B1*: the 100 smallest exceptional cases, with $a(m)$ ranging from 1.00×10^8 to 1.52×10^8, providing the first independent minimality check for exceptional values.
   - *Group B2*: 150 randomly selected non-exceptional cases ($a(m) < 10^8$, seed = 42).
   - *Group C*: all 40 resolved cases with $a(m) > 10^9$ from the initial computation, including $m = 14\,740$ ($a(m) = 1{,}994{,}463{,}433$) and $m = 16\,167$ ($a(m) = 1{,}881{,}095{,}843$), the two largest values in that dataset.
   - *Group E1*: the 4 extended-search cases with the largest Cramér ratios ($m \in \{22\,802,\ 23\,995,\ 53\,963,\ 82\,300\}$, $a(m) \in [4.4 \times 10^{10},\ 8.2 \times 10^{10}]$), with individual run times of 1 736–2 813 seconds.
   - *Group E2*: 15 uniformly sampled extended-search cases with $a(m) \in [2.0 \times 10^9,\ 7.9 \times 10^9)$, and *Group E3*: all 9 extended-search cases with $a(m) \in [1.1 \times 10^{10},\ 3.0 \times 10^{10}]$. Together these cover the full range of the extended-search dataset below the Group E1 threshold.

   In addition, a smoke-test case ($m = 11\,924$, $a(m) = 6\,119\,832\,581$, elapsed 149 s) was run as a reference check before each group E1–E3 campaign. All 319 cases were confirmed without discrepancy. The `verify_min_ext.c` program uses `__uint128_t` arithmetic for the divisibility check to avoid overflow for $p_n^2 \sim 10^{25}$ at the largest extended-search values.

In total, 10 925 individual verifications were performed across all three methods: 10 000 cross-consistency checks (Method 1), 606 SymPy divisibility checks (Method 2), and 319 independent sieve minimality checks (Method 3), with zero discrepancies.

4 Results

4.1 Summary

For $m = 1, \ldots, 120\,000$, we determined $a(m)$ in 119 995 cases: 119 931 in the initial computation (Phase 6, bound $n \leq 2 \times 10^9$) and 64 additional cases in the extended search (bound $n \leq 2 \times 10^{11}$; see Section 3.3). The remaining 5 values are listed and discussed in Section 7. Table 2 lists the ten largest values of $a(m)$ found.

We call a resolved m *exceptional* if $a(m) > 10^8$. This is an author-defined cutoff chosen for exposition; it is approximately 1300 times the median value of $a(m)$ ($\approx$ 79 000, computed from the 119 995 resolved values). There are $E_{\text{res}}(1.2 \times 10^5) = 606$ such resolved exceptional cases (542 from the initial computation plus all 64 cases resolved in the extended search, each of which has $a(m) > 2 \times 10^9$). Since the 5 unresolved values all satisfy $a(m) > 2 \times 10^{11}$ by definition of the final search bound, the true exceptional count satisfies $E(1.2 \times 10^5) \geq 611$, where

$$E(M) \;=\; E_{\text{res}}(M) + \#\big\{m \leq M : m \text{ unresolved}\big\}$$

denotes the total exceptional count; the quantity $E_{\text{res}}(M)$ counts only resolved exceptional cases throughout this paper.

The complete dataset (all 119 995 resolved values and the list of 5 unresolved m) is publicly available; see Section 8.

Table 2 Ten largest values of $a(m)$ for $m \leq 120\,000$. The Cramér-type ratio is $C(m) = a(m)/(m \log m)$ (natural logarithm throughout). All entries were found in the extended search (Section 3.3).

m	$a(m)$	p_m	$a(m)/(m \log m)$
82 300	81 961 045 941	1 052 459	87 990
53 963	50 362 759 647	665 179	85 653
23 995	47 572 503 750	274 453	196 577
22 802	44 000 698 448	259 577	192 303
37 253	29 834 694 588	443 873	76 088
48 834	23 630 637 176	596 419	44 821
37 949	23 612 836 020	453 181	59 012
43 297	20 932 270 152	523 007	45 285
69 546	17 485 936 480	876 191	22 550
109 857	14 929 726 016	1 438 973	11 709

4.2 Outstanding cases

We list the five largest Cramér-type ratios $C(m) = a(m)/(m \log m)$ among resolved $m \leq 120\,000$ in Table 3. Since $p_m \sim m \log m$ by the prime number theorem (see [6], Ch. 18), the ratio $C(m)$ is approximately $a(m)/p_m$, normalising the solution index by the size of the prime p_m. Under Cramér's probabilistic model for primes [11], treating primes as independent random events with local density $1/\log n$ near n, the expected index at which one encounters a prime satisfying a given congruence condition of modulus of order m is of order m; hence $C(m)$ measures how many times larger the actual solution is compared to this heuristic baseline. The extended search dramatically reshuffled the ranking: $m = 23\,995$ now holds the global resolved maximum ($C \approx 196\,577$), far exceeding the previous record holder $m = 4703$ ($C \approx 19\,111$). All five entries in the new top 5 were resolved in the extended search.

Table 3 Top five resolved values by Cramér-type ratio $a(m)/(m \log m)$. All entries were found in the extended search.

Rank	m	$a(m)$	p_m	$a(m)/(m \log m)$
1	23 995	47 572 503 750	274 453	196 577
2	22 802	44 000 698 448	259 577	192 303
3	82 300	81 961 045 941	1 052 459	87 990
4	53 963	50 362 759 647	665 179	85 653
5	37 253	29 834 694 588	443 873	76 088

For $m = 23\,995$: $p_m = 274\,453$, $a(m) = 47\,572\,503\,750$, $s = m + a(m) = 47\,572\,527\,745$. The divisibility and minimality of this value were independently confirmed by `verify_min_ext.c` (Group E1, elapsed 1 854 seconds). For $m = 11\,924$ (previous resolved record, now outside the top 5): $p_m = 127\,289$, $a(m) = 6\,119\,832\,581$, $s = m + a(m) = 6\,119\,844\,505$, $C \approx 54\,679$; independently verified by both methods.

4.3 Growth of resolved exceptional cases

Let $E_{\text{res}}(M)$ denote the number of resolved values $m \leq M$ with $a(m) > 10^8$. Table 4 shows the growth of $E_{\text{res}}(M)$ over successive ranges of 10 000.

Table 4 Count of resolved exceptional cases $a(m) > 10^8$ by range of m (including extended-search results).

Range of m	Count
1–10 000	2
10 001–20 000	30
20 001–30 000	25
30 001–40 000	30
40 001–50 000	44
50 001–60 000	52
60 001–70 000	50
70 001–80 000	63
80 001–90 000	67
90 001–100 000	80
100 001–110 000	81
110 001–120 000	82
Total	606

5 Statistical Analysis

5.1 Heavy tail

The distribution of $a(m)$ for resolved $m = 1, \ldots, 120\,000$ is strongly heavy-tailed. We apply the Hill estimator [4] as an exploratory diagnostic tool; since the data are deterministic and not i.i.d., this should be understood as descriptive rather than as a formal inference about an underlying distribution (see [13] for a discussion of the limitations of power-law fitting in empirical data). Given N_s order statistics $X_{(1)} \leq \cdots \leq X_{(N_s)}$, the Hill estimator with threshold parameter k is defined as

$$\hat{\alpha}_{\text{Hill}}(k) = \left(\frac{1}{k} \sum_{i=1}^{k} \log \frac{X_{(N_s-i+1)}}{X_{(N_s-k)}} \right)^{-1}.$$

The Hill statistic, computed over the top 10% of the 119 931 values from the initial computation ($k = 11\,993$, threshold $a(m) > 1\,364\,314$), takes the value

$$\hat{\alpha}_{\text{Hill}} = 0.713.$$

Standard confidence intervals for the Hill estimator assume i.i.d. data and do not apply to this deterministic dataset; the value above should be interpreted as a descriptive summary only. The 64 values from the extended search are all in the extreme tail ($a(m) > 2\times10^9$) and represent 0.05% of the dataset; their inclusion does not materially change the Hill estimate (the statistic is driven by the top 10% of the initial dataset, and 64 additional points in the far tail shift $\hat{\alpha}$ by less than 0.002). The statistic varies with threshold:

$$\hat{\alpha}(1\%) = 0.817, \quad \hat{\alpha}(5\%) = 0.732, \quad \hat{\alpha}(10\%) = 0.713, \quad \hat{\alpha}(20\%) = 0.676.$$

The 10% threshold is adopted as the primary estimate, balancing sensitivity to the tail against stability of the statistic. The threshold-dependence indicates that the tail is not a pure power law; the values of $\hat{\alpha}_{\text{Hill}}$ should be interpreted only as an indication of heavy-tailed behaviour, not as evidence of a specific parametric form.

5.2 Cramér-type ratio

The Cramér-type ratio $C(m) = a(m)/(m \log m)$, as defined in Section 4.2, has median over resolved $m = 2, \ldots, 120\,000$ (computed from the 119 995 resolved values)

$$\widetilde{C} \approx 0.154,$$

stable and well-defined across the full range; the 64 newly resolved values lie far in the upper tail and do not affect the median.

6 Heuristic Model and Comparison with Data

6.1 Density argument

We present below a density-based heuristic for the expected scale of $a(m)$; as the data will show, the model fails quantitatively to predict the observed scale, indicating the need for a refined understanding.

Fix m and consider the search for a valid n. For a given candidate $s = m + n$, a solution exists if and only if p_n falls in one of the $2^{\omega(s)}$ residue classes of the form $r \cdot p_m \pmod{s}$, where r ranges over the square roots of $-1 \pmod{s}$. By Dirichlet's theorem [6], the asymptotic density of primes in each such class is $\varphi(s)^{-1}$ for fixed s, where $\varphi(s)$ denotes Euler's totient function (the number of integers in $\{1, \ldots, s\}$ coprime to s); we note that applying this for moduli s that grow up to $\sim 2 \times 10^9$ while checking primes up to $\sim 4.7 \times 10^{10}$ goes beyond the range of standard uniformity results (Siegel–Walfisz, see [6], Ch. 21), and the following heuristic should be treated as qualitative. The total proportion of admissible residue classes modulo s is

$$D(s) \;=\; \frac{2^{\omega(s)}}{\varphi(s)}$$

for odd s; for $s \equiv 2 \pmod{4}$ the numerator is $2^{\omega(s)-1}$, a distinction absorbed into the leading constant in the estimates below. Note that $D(s)$ is the *proportion of admissible residue classes*, not a probability of success for a randomly chosen prime near s; the following derivation should be regarded as a rough heuristic envelope rather than a rigorous estimate.

Using the normal-order estimate $\omega(s) \approx \log\log s$ [8] and the extremal lower bound $\varphi(s) \gtrsim s \cdot e^{-\gamma}/\log\log s$ [9] (the first is a normal-order result valid for almost all s, the second an extremal bound realised at primorials; their combination has no rigorous standing), one obtains:

$$D(s) \;\lesssim\; \frac{(\log s)^{\log 2} \cdot e^{\gamma} \log\log s}{s}.$$

Treating each candidate n as independently selected (in the spirit of the Cramér model, which treats primes as random with local density $1/\log n$) with probability $D(m+n)$ of yielding a valid solution, the expected value of $a(m)$ satisfies $\sum_{n=1}^{N} D(m+n) \approx 1$, which for slowly varying D reduces to $\int_m^{m+N} D(s)\,ds \approx 1$. Substituting the bound on $D(s)$ and integrating yields

$$\int_m^{m+N} \frac{(\log s)^{\log 2} \cdot e^{\gamma} \log\log s}{s}\,ds \;\approx\; e^{\gamma} \cdot (\log m)^{\log 2} \cdot \log\log m \cdot \frac{N}{m} \;\approx\; 1,$$

where the approximation treats $\log s \approx \log m$ and $\log\log s \approx \log\log m$ over the range $[m, m+N]$, valid since $N \ll m$ in the regime of interest. This gives $N \sim m/(e^{\gamma}(\log m)^{\log 2} \log\log m)$; for $m = 10^5$ this yields $\approx 0.04\,m$ — well below the observed median $a(m) \approx 1.3\,m$. This large discrepancy shows that the density model

does not explain the observed scale of $a(m)$. This prediction is purely illustrative and should not be interpreted as a rigorous asymptotic estimate; the underlying derivation combines a normal-order result with an extremal bound in a mathematically unjustified way, as noted above. The Cramér-type normalization $m \log m$ is motivated independently by the prime-index analogy (Section 5.2), not derived from $D(s)$.

6.2 Comparison with data

The density model provides a qualitative explanation for the heavy tail (the factor $(\log s)^{\log 2}$ in $D(s)$ produces slower-than-power-law decay), but fails quantitatively: it predicts the typical scale of $a(m)$ as $\sim m/(e^{\gamma}(\log m)^{\log 2} \log\log m) \approx 0.04\, m$, whereas the typical $a(m)$ scales as approximately $1.3\, m$. The growing count $E_{\mathrm{res}}(M)$ in Table 4 is qualitatively consistent with increasing exceptional-case frequency, but the growth factor of 303 from $E_{\mathrm{res}}(10^4) = 2$ to $E_{\mathrm{res}}(1.2 \times 10^5) = 606$ is far larger than any simple log-power formula would predict. We therefore regard the density model as providing only a qualitative motivation for heavy-tailed behaviour, and do not propose a quantitative formula for $E_{\mathrm{res}}(M)$.

6.3 Sources of discrepancy

The density model of Section 6.1 underestimates the typical value of $a(m)$ by a factor of approximately 30 (predicting $\sim 0.04\, m$ versus observed $\sim 1.3\, m$). We identify three structural sources of this discrepancy, none of which is captured by the independence assumptions underlying the model.

Correlated candidates.

The model treats each candidate n as an independent trial. In reality, the moduli $s = m + n$ and $s' = m + n + 1$ are consecutive integers, and their divisibility by small primes is strongly correlated: if $q \mid s$ then $q \nmid s'$. Over a window of q consecutive values of s, exactly one is divisible by q; this short-range anticorrelation means that the "success events" (finding a prime in an admissible residue class) are not independent, and the waiting-time distribution differs from the geometric distribution assumed by the model.

Varying modulus.

For each candidate n, the modulus $s = m + n$ changes, so the set of admissible residue classes changes with n. The density model averages over s as though the residue class condition were fixed, but in practice the target moves with every step. In particular, $D(s)$ fluctuates between zero (when s has a prime factor $q \equiv 3 \pmod 4$) and $\gg 1/s$ (when s is a product of small primes $\equiv 1 \pmod 4$); this intermittency produces longer runs of consecutive failures than the averaged model predicts.

Interaction with p_m.

The density $D(s)$ accounts for the proportion of admissible residue classes modulo s, but the actual condition also requires p_n to be a prime landing in one of these

classes. The distribution of primes in residue classes modulo s is governed by Dirichlet's theorem, which gives equidistribution only asymptotically; for the finite ranges involved ($p_n \leq 4.7 \times 10^{10}$ at bound $n = 2 \times 10^9$, while s can be as large as $\sim 2 \times 10^9$), there can be significant deviations from equidistribution, particularly for moduli s with special arithmetic properties relative to p_m.

These three effects compound to produce a systematic underestimate of the waiting time. Collectively, they plausibly explain much of the observed factor of approximately 30 between the heuristic prediction ($\sim 0.04\,m$) and the data ($\sim 1.3\,m$). A refined model incorporating even the first correction (correlated candidates via an inclusion-exclusion over small primes) would likely close much of the gap; see also [16] for a discussion of corrections to the classical Cramér heuristic in related contexts. We leave the development of such a model as a direction for future work.

The outstanding Cramér-type ratio of $m = 23\,995$ (ratio $\approx 196\,577$, the global resolved maximum over $m \leq 120\,000$) may reflect an especially unfavourable interaction between $p_{23\,995} = 274\,453$ and the moduli $s = m + n$, producing a suppression of $D(s)$ not captured by the average model. Identifying the specific arithmetic mechanism responsible for such extreme outliers remains an open question.

7 Unresolved Cases

The 69 values of $m \leq 120\,000$ that were unresolved in the initial computation ($n \leq 2 \times 10^9$) were subjected to the extended search described in Section 3.3. Of these, 64 were resolved, leaving 5 values unresolved within the final search bound $n \leq 2 \times 10^{11}$. They are listed in Table 5, together with p_m, the filter-detected obstruction rate, and estimated lower bounds on the Cramér ratio $C(m)$. The *obstruction rate* is the percentage of candidate values of n (within the search range) eliminated by the modular filter of Section 3.1 before any arithmetic is performed.

Table 5 The 5 unresolved values of $m \leq 120\,000$, with p_m, filter-detected obstruction rate (percentage of candidates eliminated by the modular filter; see Section 3.1), and estimated Cramér ratio lower bound. No solution was found with $n \leq 2 \times 10^{11}$.

m	p_m	Obstr. (%)	$C(m) >$
37 249	443 837	76.2	510 126
66 257	831 301	76.9	271 910
76 868	977 087	77.0	231 280
98 379	1 276 747	77.2	176 831
117 862	1 552 879	77.4	145 316

All 5 cases exhibit obstruction rates in the range 76.2–77.4%, consistent with the 75.7–77.4% range observed for the original 69 unresolved cases. The extended search resolved 64 of those 69 cases (Table 1), demonstrating that the obstruction rate alone

does not determine whether a case is resolvable — rather, it is the Cramér ratio that correlates with difficulty. The 5 remaining cases are precisely those with the highest estimated Cramér ratios among the original 69, all exceeding 145 000.

Resolution curve.

The extended search resolved cases at a strongly decreasing rate: 51 cases in the first range $n \in [2 \times 10^9, 10^{10}]$, 9 more in $[10^{10}, 3 \times 10^{10}]$, 2 in $[5 \times 10^{10}, 10^{11}]$, and 0 in the final range $[10^{11}, 2 \times 10^{11}]$. Two consecutive sessions (S3 and S4) resolved no cases.

Computational lower bounds.

The exhaustive search establishes the following.

Theorem 7.1 (Computational lower bounds) *For each* $m \in \{37\,249,\ 66\,257,\ 76\,868,\ 98\,379,\ 117\,862\}$*, the divisibility condition* $(m+n) \mid p_m^2 + p_n^2$ *has no solution with* $1 \le n \le 2 \times 10^{11}$*. Consequently, either* $a(m) > 2 \times 10^{11}$ *or Conjecture 1.1 fails for this value of* m*.*

Proof The search program `sun_ext_v11.c` tests, for each of the five values of m, every integer n in the range $1 \le n \le 2 \times 10^{11}$: it enumerates primes p_n via a segmented sieve of Eratosthenes and, for each n not eliminated by the modular filter of Section 3.1, evaluates the divisibility condition $(p_m^2 + p_n^2) \bmod (m+n) = 0$ directly using 128-bit arithmetic (Section 3.3). The modular filter is *sound* (Proposition 2.1): it eliminates only values of n for which no solution can exist, so no valid candidate is skipped. For all five values of m, the search returns no solution in the stated range; the sieve is deterministic and complete, so the conclusion follows. The source code is publicly available [20]. □

Remark 7.2 The result of Theorem 7.1 is conditional on the correctness of the deterministic search implementation. The implementation has been independently spot-checked for 319 values via Method 3 of Section 3.5, with zero discrepancies; this includes the 4 resolved cases with the largest Cramér ratios.

In particular, the Cramér-type ratio for each of these values satisfies $C(m) = a(m)/(m \log m) > 2 \times 10^{11}/(m \log m)$, yielding the lower bounds in Table 5.

Computational challenge.

We propose the five values of Theorem 7.1 as computational challenge instances: for each, determine $a(m)$ or establish that no solution exists. The decreasing resolution rate (Table 1) and the absence of any resolution in the range $n \in [10^{11}, 2 \times 10^{11}]$ indicate that resolving these cases will require searching substantially beyond $n = 2 \times 10^{11}$. Whether any of these values constitutes a genuine counterexample to Conjecture 1.1 remains an open question; the pattern of progressive resolution observed in this study (from 69 unresolved at bound 2×10^9 to 5 at bound 2×10^{11}) is consistent with these being extreme outliers of the tail distribution rather than structural failures of the conjecture.

8 Conclusion

We have determined $a(m)$ for 119 995 values of $m \leq 120\,000$, with 5 values remaining unresolved within our final search bound $n \leq 2 \times 10^{11}$. The initial computation ($n \leq 2 \times 10^9$) resolved 119 931 cases; a targeted extended search resolved 64 of the 69 initially unresolved cases, reducing the count to 5. The resolved dataset reveals a heavy-tailed distribution with Hill statistic $\hat{\alpha}_{\text{Hill}} \approx 0.713$, a growing density of exceptional cases, and a new outstanding resolved case at $m = 23\,995$ whose Cramér-type ratio ($\approx 196\,577$) is the global resolved maximum over the entire range studied, more than ten times the previous record holder $m = 4703$ ($\approx 19\,111$).

We proved that the modular filter eliminates an asymptotically increasing proportion of candidates (Proposition 3.1), with the theoretical rate for the truncated filter ($\approx 78.5\%$) consistent with the observed range (73–77%). We identified three structural sources for the quantitative failure of the density-based heuristic (Section 6.3): correlated candidates, varying moduli, and finite-range deviations from equidistribution in arithmetic progressions.

For the 5 unresolved cases, we established computational lower bounds $a(m) > 2 \times 10^{11}$ (Theorem 7.1) and proposed them as computational challenge instances. The progressive resolution of the original 69 unresolved cases — from 69 at bound 2×10^9 to 5 at bound 2×10^{11} — demonstrates that the conjecture was verified computationally for all but the most arithmetically resistant values of m in this range.

The heuristic density model provides a qualitative explanation for the heavy tail but fails quantitatively to predict the scale of $a(m)$ or the growth rate of exceptional cases.

Comparison with Conjecture 4.3(i).

A companion computation [19] applies the same methodology to sequence A248044 [18], which encodes Sun's Conjecture 4.3(i): $(m+n) \mid \pi(m)^2 + \pi(n)^2$. Computing $a(m)$ for $m \leq 100\,000$ with bound $n \leq 10^{13}$ resolves 99 998 of 100 000 cases, leaving only 2 unresolved. The comparison reveals instructive structural differences: the obstruction rate is lower ($\sim 57\%$ vs. 73–77% here), the maximum resolved Cramér-type ratio is much larger ($\sim 6.5 \times 10^7$ vs. $\sim 1.97 \times 10^5$), and the two unresolved cases form an arithmetically structured cluster sharing $\pi(m) = 2225 = 5^2 \times 89$, whose factorisation into primes $\equiv 1 \pmod 4$ produces maximal modular obstruction — a phenomenon absent in the present sequence, where p_m is always prime and the obstruction structure is more uniform. These contrasts indicate that the difficulty of Sun's divisibility conjectures depends critically on the arithmetic function involved (p_k vs. $\pi(k)$), not only on the shared divisibility framework $(m+n) \mid f(m)^2 + f(n)^2$.

Future work.

Natural directions include an extended computational search for the 5 remaining unresolved values of A247975 (and the 2 of A248044), a refined probabilistic model incorporating the correlation structure identified in Section 6.3, and a systematic comparative study across the full family of Sun's divisibility conjectures [1].

Acknowledgements. The author thanks Zhi-Wei Sun for the beautiful conjecture that motivated this work, and Chai Wah Wu, whose extension of the OEIS b-file for A247975 to $m \leq 10\,000$ provided a valuable cross-check for our results.

Declarations

Funding No funding was received for conducting this study.
Competing interests The author has no relevant financial or non-financial interests to disclose.
Data availability The complete dataset (119 995 resolved values and 5 unresolved m-values) is publicly available at:

- GitHub repository: https://github.com/carcorti/A247975
- Permanent Zenodo archive (DOI): https://doi.org/10.5281/zenodo.18920371
- OEIS: https://oeis.org/A247975

The repository will be synchronized with the submitted version of the manuscript and all associated data files prior to publication.
Code availability The C source code for the main search (`sun_v6.c`), the extended-search program (`sun_ext_v11.c`), the independent verification sieves (`verify_min.c` and `verify_min_ext.c`), and the Python/SymPy verification scripts (`verify_ext_sympy.py`, `run_verify_ext_batch.py`) are available at the GitHub and Zenodo repositories listed above.
Author contribution Not applicable (single author).
AI assistance The author used Claude (Anthropic) as an AI assistant during the preparation of this manuscript. The AI assisted with code review, verification of computational results, and editorial drafting of the manuscript. All scientific content, mathematical proofs, computational results, and final decisions regarding the text are the sole responsibility of the author.

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